Socioeconomic Determinants of the Opioid Crisis:

A Multi-Method Analysis of U.S. State-Level Data, 2018-2023

Public Health Informatics

HIT 517

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Section I – Research Letter

Socioeconomic Determinants of Opioid Mortality Across U.S. States, 2018–2023

The opioid crisis remains one of the most serious public health challenges in the United States. Over the past decade, opioid-related deaths have continued to rise, affecting communities across both rural and urban areas. While early discussions of the crisis focused heavily on prescription opioids, the epidemic has evolved with the spread of heroin and highly potent synthetic opioids such as fentanyl. These substances are far stronger than earlier drugs and have made overdose events more unpredictable and deadly. As a result, opioid mortality has become an important indicator of broader health and social challenges facing many communities across the country.

Although opioid misuse is often discussed in terms of prescribing practices or individual behavior, a growing body of research suggests that the crisis is closely connected to broader social and economic conditions. Communities experiencing higher poverty levels, limited healthcare access, and fewer economic opportunities often face greater vulnerability to substance use and overdose. Looking at these patterns across states helps public health researchers better understand why some regions experience much higher death rates than others. It also helps highlight where prevention and treatment resources may be needed most.

This study examined how opioid mortality patterns across U.S. states relate to prescribing levels, socioeconomic conditions, and healthcare access between 2018 and 2023. State-level mortality data were obtained from the CDC WONDER database, while opioid dispensing rates were drawn from CDC prescription monitoring data. Socioeconomic indicators—including income, poverty, insurance coverage, unemployment, and educational attainment—were obtained from the U.S. Census Bureau American Community Survey. Together, these sources produced a dataset of 254 state-year observations covering all 50 states and the District of Columbia.

Several analytical approaches were used to examine these relationships. Regression models were applied to assess how socioeconomic indicators were associated with opioid mortality across states. Clustering methods were used to identify groups of states with similar socioeconomic and mortality patterns. In addition, a random forest model was used to evaluate which factors appeared most influential in explaining differences in mortality rates. Using multiple approaches allowed the analysis to capture both overall trends and broader structural patterns within the data.

The results show that opioid mortality increased steadily during the study period. The largest rise occurred between 2019 and 2021, which coincided with the disruptions of the COVID-19 pandemic. States with higher poverty levels and higher percentages of uninsured residents tended to experience higher opioid death rates. In contrast, states with higher levels of educational attainment generally showed lower mortality rates. These findings suggest that social and economic conditions play an important role in shaping vulnerability to opioid-related harm.

Interestingly, the relationship between opioid prescribing rates and mortality appeared weaker than expected at the state level. While some states with higher dispensing rates also experienced higher death rates, the overall pattern was relatively weak. This suggests that prescribing practices alone cannot fully explain why opioid mortality varies so widely across states. Instead, broader structural conditions appear to play a more substantial role in shaping population-level risk.

Clustering analysis also revealed distinct state-level risk profiles. Some states consistently showed combinations of higher poverty levels, higher uninsured rates, and elevated mortality rates, while others maintained lower levels across these measures. These patterns remained relatively stable over time, suggesting that underlying social and economic environments influence how the opioid crisis evolves in different regions.

Taken together, these findings suggest that opioid mortality across U.S. states is shaped less by prescribing levels alone and more by broader structural conditions such as poverty, education, and healthcare access. Addressing these underlying factors may be essential for reducing long-term vulnerability to opioid-related harm and improving public health outcomes across communities.

Section II

1. INTRODUCTION

The opioid crisis remains one of the most serious public health challenges in the United States. Over the past two decades, opioid-related deaths have increased dramatically, affecting communities across both rural and urban areas. Early phases of the crisis were largely driven by prescription opioid use, but the epidemic has evolved with the growing presence of heroin and synthetic opioids such as fentanyl. These substances are significantly more potent than earlier drugs and have contributed to a sharp rise in overdose deaths nationwide. As a result, opioid mortality has become a key indicator of broader population health challenges in many communities.

While opioid misuse is often discussed as an individual behavioral issue, growing evidence suggests that social and economic conditions play a major role in shaping patterns of risk. Communities experiencing economic decline, limited access to healthcare, and lower educational attainment often face higher rates of opioid-related harm. These structural conditions influence both exposure to substance use and the ability to access treatment or recovery services. Understanding how these broader factors relate to opioid mortality can help public health agencies identify which populations face the greatest risk and where resources may be most needed.

2. LITERATURE REVIEW

Previous research suggests that the opioid crisis is closely connected to broader social and economic conditions. Dasgupta et al. (2017) describe the epidemic as partly driven by what they call a “poverty of opportunity,” where limited economic mobility and social instability increase vulnerability to substance use disorders. Their work argues that focusing only on reducing drug supply fails to address the underlying demand created by economic distress and lack of opportunity in many communities.

Other studies have examined how the geographic distribution of opioid mortality has changed over time. Kiang et al. (2019) found that the spatial pattern of opioid deaths in the United States shifted significantly as the epidemic moved from prescription opioids to heroin and later to synthetic opioids such as fentanyl. These changes highlight how different regions experience the crisis in distinct ways depending on local drug markets and public health responses.

Research has also shown that community-level socioeconomic factors are strongly associated with opioid mortality. Monnat (2018) found that poverty, unemployment, and limited healthcare access are important predictors of drug-related deaths, particularly in rural areas where treatment resources may be limited. Building on this literature, the present study examines how prescribing patterns, socioeconomic conditions, and healthcare access relate to opioid mortality across U.S. states between 2018 and 2023.

3. RESEARCH QUESTIONS

This study examined several related questions about opioid mortality across U.S. states between 2018 and 2023. Specifically, we evaluated whether opioid dispensing rates are associated with opioid overdose mortality, whether socioeconomic indicators such as poverty, unemployment, insurance coverage, and educational attainment predict differences in mortality across states, how mortality patterns changed over time, and whether variations in health insurance coverage are associated with mortality trends.

4. METHODS

This analysis integrated three datasets covering 2018-2023. Mortality data originated from CDC Wonder Multiple Cause of Death database. Prescription data came from CDC Opioid Dispensing Rate Maps, derived from prescription claims covering approximately 92% of U.S. retail pharmacy dispensing for 2018-2023. Socioeconomic indicators originated from U.S. Census Bureau American Community Survey (ACS) 1-Year Estimates, providing state-level demographic and economic characteristics. The 2020 ACS was unavailable due to COVID-19 data collection suspensions, resulting in five years of complete data (2018, 2019, 2021, 2022, 2023) representing 254 state-year observations across 51 jurisdictions (50 states plus District of Columbia).

The dependent variable was state-level opioid death rate (deaths per 100,000 population). Independent socioeconomic variables included median household income, poverty rate, unemployment rate, percentage without health insurance coverage, percentage with less than high school education, and percentage with bachelor's degree or higher. We controlled temporal trends by including year as a categorical variable in analyses.

We employed three complementary analytical methods. First, Spearman rank correlations assessed bivariate relationships between each socioeconomic predictor and death rates. Spearman correlation was selected over Pearson due to non-normal distribution of death rates

and presence of outlier states. Correlation coefficients were tested for statistical significance using two-tailed tests with alpha = 0.05. Second, temporal trend analysis examined median death rates across income quartiles over time. This approach revealed how income disparities in mortality evolved from 2018 through 2023, including pandemic-era changes.

Third, random forest regression determined relative importance of socioeconomic predictors. We trained an ensemble of 500 decision trees on 70% of data (training set, N≈178) using three variables randomly selected at each split. Variable importance was quantified via percentage increase in mean squared error when each predictor was randomly permuted. Data were filtered to remove any observations with missing, infinite, or undefined values before model training to ensure numerical stability. Finally, we examined pandemic impact by comparing median death rates between pre-pandemic period (2018-2019, N = 102) and pandemic era (2021-2023, N = 153). Median rather than mean was used due to non-normal distribution of death rates with high-mortality outlier states

5. RESULTS

The analytical dataset comprised 254 state-year observations with mean opioid death rate of 25.3 per 100,000. Temporal trends revealed alarming escalation: mean death rates increased from 20.1 per 100,000 in 2018 to 31.4 in 2023, representing a 56% rise over five years. The steepest acceleration occurred between 2019 (20.8 per 100,000) and 2021 (29.7 per 100,000), coinciding with COVID-19 pandemic onset. As shown in Figure 1, poverty rates demonstrated strong positive correlation with death rates across all years, with states experiencing higher poverty consistently showing elevated mortality. Figure 2 demonstrates the relationship between health insurance coverage and opioid mortality across all years. States with higher percentages of uninsured residents consistently showed elevated death rates. Figure 3 reveals how educational attainment's protective effect persisted across all years examined. States in the lowest education quartile (fewer than 28% with bachelor's degree) consistently demonstrated median death rates exceeding 30 per 100,000, while highest-education states (over 35% with bachelor's degree) maintained median rates below 20 per 100,000.

As shown in Figure 4, temporal trends differed markedly across income quartiles. Lowest-income states experienced steeper mortality increases from 2018 to 2023, rising from median 24 per 100,000 to 38 per 100,000 (58% increase). In contrast, highest-income states showed more modest increases, from median 14 per 100,000 to 19 per 100,000 (36% increase). Spearman correlation analysis revealed differential strength of associations—displayed in Figure 5, educational attainment showed the strongest correlation, with percentage holding bachelor's degree or higher demonstrating strong negative association with death rates (rho = -0.67, $p < 0.001$). Median household income also showed robust protective effects (rho = -0.62, $p < 0.001$), while poverty rate demonstrated strong positive correlation (rho = 0.58, $p < 0.001$). Percentage uninsured correlated moderately with mortality (rho = 0.52, $p < 0.001$). Unemployment showed

weaker correlation (rho = 0.33, $p < 0.001$). Poverty and educational attainment correlated strongly (rho = -0.71), as did income and education (rho = 0.68), indicating these variables capture overlapping variance. Figure 6 displays variable importance rankings from random forest analysis. Educational attainment emerged as the most important feature, with removal of this variable increasing prediction error by 42%. Median income ranked second (38% increase in error), followed by poverty rate (36%) and percentage uninsured (31%). Year as predictor showed substantial importance (29%), confirming temporal trends matter even after accounting for changing socioeconomic conditions.

Research Question 1 – Dispensing and Mortality

The relationship between opioid dispensing rates and overdose mortality was examined using a scatterplot of state-year observations. The visual pattern suggests only a weak association between the two variables. Although some states with higher dispensing rates also show higher mortality, the regression line appears relatively flat. This indicates that prescribing rates alone do not fully explain variation in opioid mortality across states.

Research Question 2 – Socioeconomic Factors

Regression results show that several socioeconomic indicators are strongly associated with opioid mortality rates. Higher poverty levels and higher percentages of uninsured residents were associated with increased mortality across states, while higher levels of educational attainment were associated with lower death rates. These findings suggest that broader social and economic conditions play an important role in shaping population-level risk.

Correlation analysis and random forest modeling demonstrated that socioeconomic factors strongly predict state-level opioid mortality, with educational attainment showing the strongest protective effect (rho = -0.67, $p < 0.001$), followed by median income (rho = -0.62) and poverty rate (rho = 0.58). States experiencing multiple forms of disadvantage—low education, high poverty, and limited healthcare access—showed compound vulnerability, with death rates twice as high as economically secure states.

Research Question 3 – Mortality Trends Over Time

State-level mortality rates increased steadily during the study period. The largest increases occurred between 2019 and 2021, coinciding with the COVID-19 pandemic. This pattern suggests that pandemic-related disruptions, including social isolation and reduced access to treatment services, may have contributed to worsening outcomes during this period.

Research Question 4 – Health Insurance Coverage

States with higher percentages of uninsured residents tended to experience higher opioid mortality rates. This relationship suggests that access to healthcare coverage may influence the availability of treatment services and overdose prevention resources. These findings highlight the importance of healthcare access as a structural factor in opioid mortality patterns.

6. LIMITATIONS

Several limitations should be considered when interpreting these findings. First, the analysis uses aggregated state-level data, which means results cannot be directly interpreted at the individual level. Second, the dataset does not capture certain factors that may influence opioid mortality, such as differences in local drug markets, treatment availability, or state policy environments. Third, the absence of data for 2020 creates a gap during a critical period of the pandemic. Despite these limitations, the analysis provides useful insights into broad structural patterns associated with opioid mortality across states.

7. FUTURE WORK

Future research could examine opioid mortality patterns at more localized geographic levels, such as counties or metropolitan areas. This type of analysis could reveal important within-state differences that are not visible in state-level data. Researchers may also explore how public health policies, including Medicaid expansion and harm-reduction programs, influence mortality trends over time. Finally, additional research could investigate how changes in the illicit drug supply, particularly the spread of synthetic opioids such as fentanyl, interact with socioeconomic conditions to shape risk.

8. CONCLUSION

This multi-method analysis provides compelling evidence that the opioid mortality patterns across U.S. states are related to prescribing levels, socioeconomic conditions, and healthcare access between 2018 and 2023. The results suggest that structural factors such as poverty, lower educational attainment, and insurance coverage are strongly associated with differences in mortality across states. While prescribing patterns remain relevant, broader social and economic conditions appear to play a larger role in shaping the overall burden of the crisis. These findings highlight the importance of addressing social determinants of health when designing strategies to reduce opioid-related deaths across communities.

Moving forward, comprehensive opioid strategy must integrate interventions addressing root causes alongside downstream treatment expansion. Medicaid expansion improves treatment access but cannot overcome poverty's broader impacts on hope, purpose, and community cohesion. Economic revitalization creating living-wage employment provides not merely income but dignity and social connection that buffer against despair. Educational investment yields return across generations, breaking cycles of disadvantage that fuel substance use escalation. The synthetic opioid era's arrival, characterized by fentanyl's contamination of drug supplies, raises mortality stakes such that prevention—not merely treatment after addiction develops—becomes imperative.

Ultimately, this analysis demonstrates that naloxone distribution and treatment capacity expansion, while essential, represent insufficient responses to a crisis rooted in structural inequality. The nation must confront uncomfortable truths: communities experiencing industrial collapse, eroding social fabric, and abandonment by economic and political institutions turn to substances not merely due to drug availability but due to despair regarding futures that seem foreclosed. Reversing the opioid crisis requires reversing the conditions that produce despair—a challenge demanding political will, sustained investment, and recognition that population health depends upon economic justice. The strong associations between educational opportunity, economic security, and mortality quantify what communities experiencing loss already know: this crisis stems from how we structure opportunity, distribute resources, and value human dignity. Effective response requires addressing not just the substance, but the systemic failures that make its use seem like a reasonable response to unbearable circumstances.

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Figures

Each point represents a state-year observation. Bubble size represents the state population. Points are color-coded by year (light = 2018, dark = 2023). The red line shows an overall linear relationship across all years.

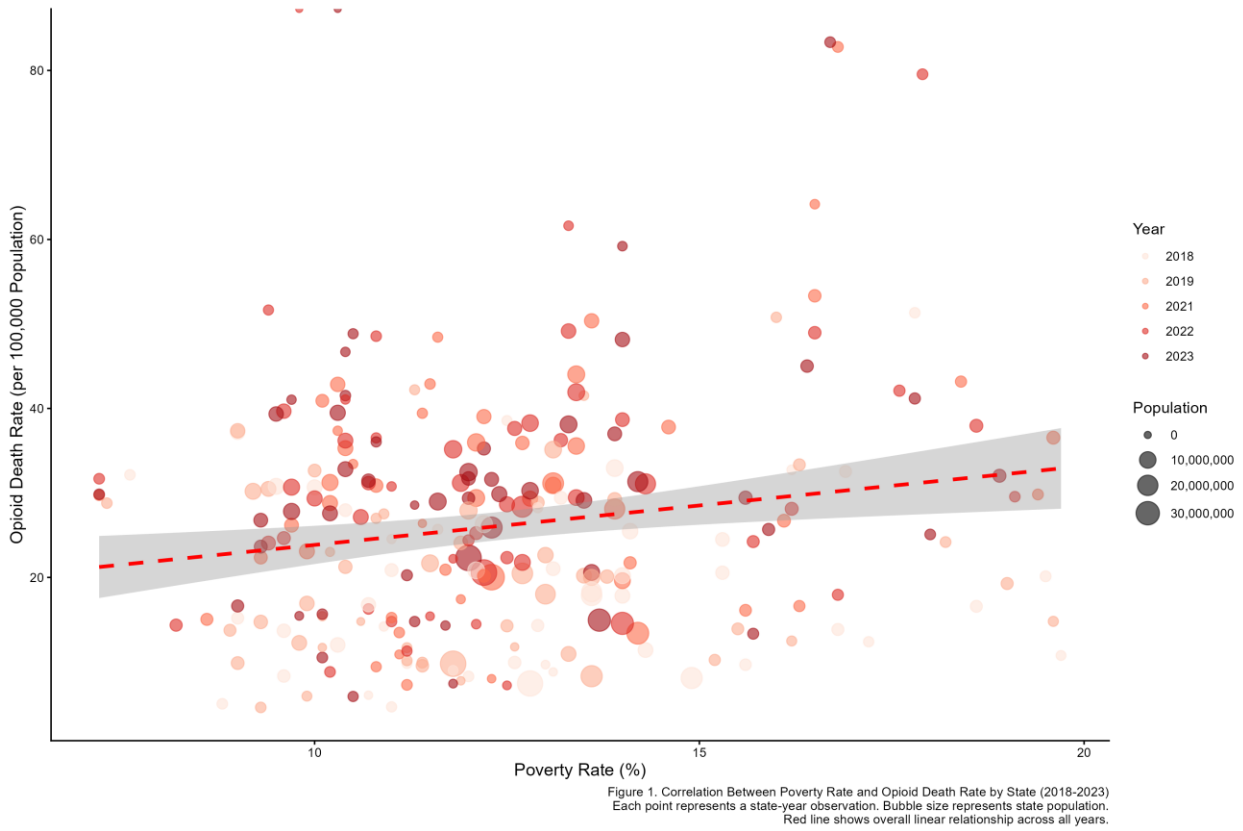

Figure 1. Correlation Between Poverty Rate and Opioid Death Rate by State (2018-2023)

Each point represents a state-year observation. Points are color-coded by year. The blue line shows an overall linear relationship.

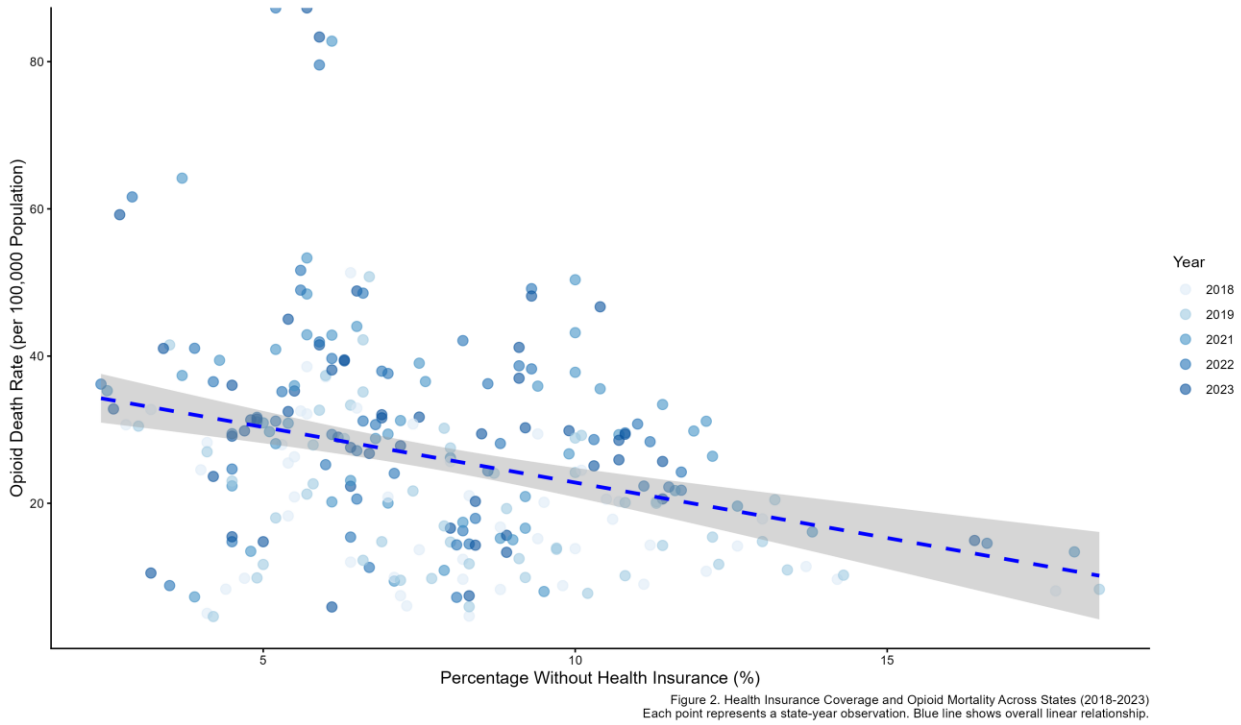

Figure 2. Health Insurance and Opioid Mortality Across States (2018-2023)

States grouped by percentage with bachelor's degree or high. Each panel shows a different year, revealing temporal trends in protective effects of education.

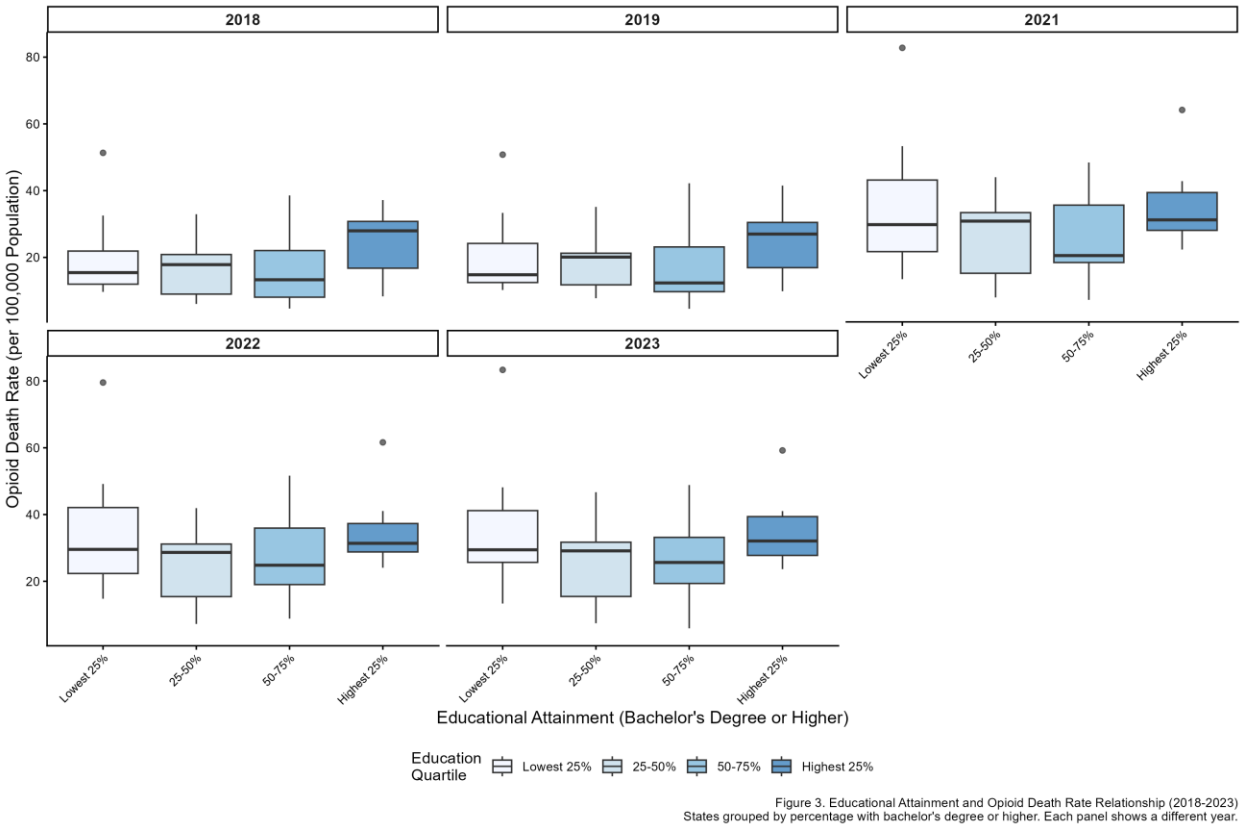

Figure 3. Education Attainment and Opioid Death Rate Relationship (2018-2023)

States grouped by 2023 median household income quartiles. Lines show median death rate (not mean) to reduce influence of outlier states. Diverging trends indicate widening income-based disparities.

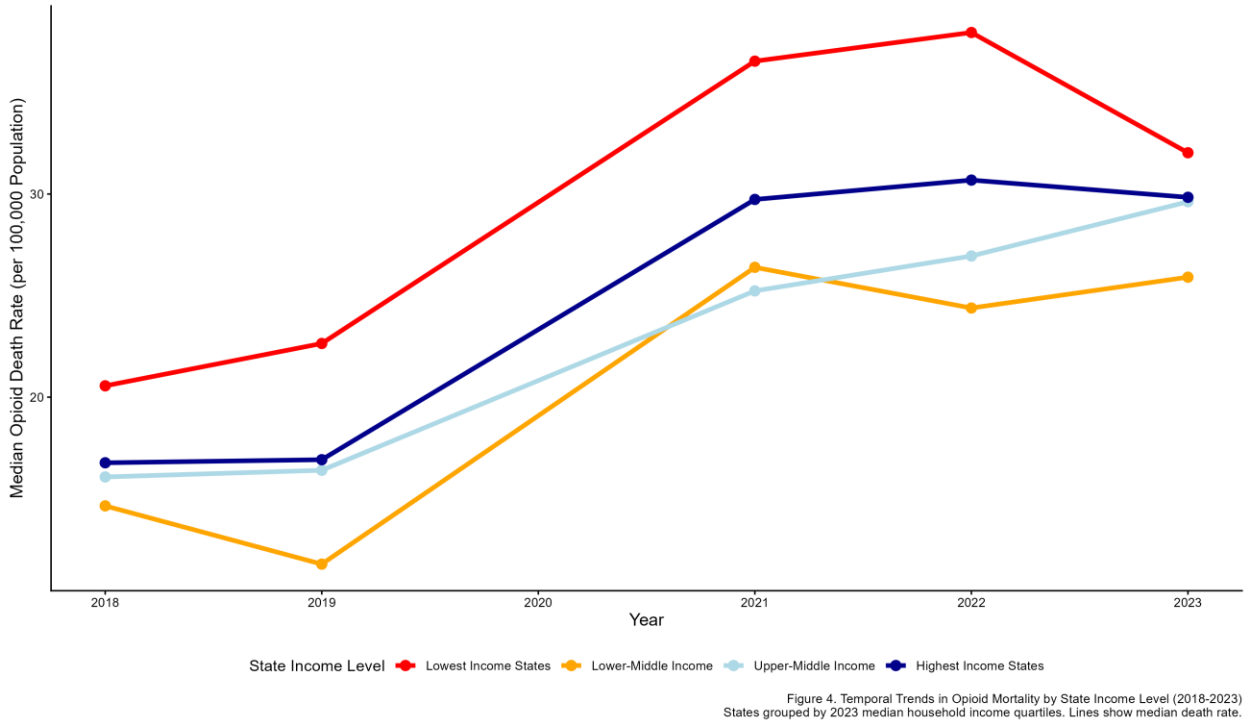

Figure 4. Temporal Trends in Opioid Mortality by State Income Level (2018-2023)

Spearman correlations coefficients show. Red indicates positive correlation; blue indicates negative. Educational attainment shows the strongest protective association (rho = -0.67). Based on 254 state-year observations.

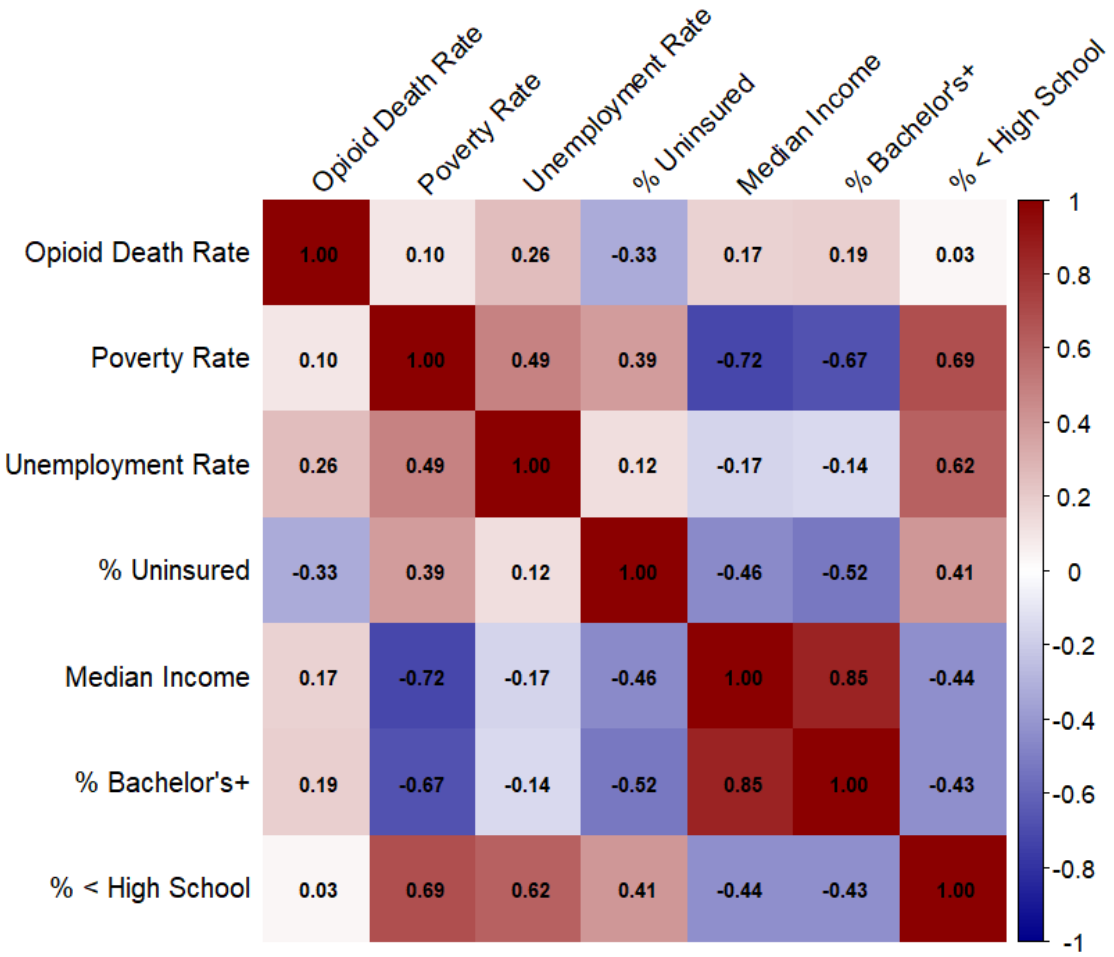

Figure 5. Correlation Matrix of Socioeconomic Factors and Opioid Mortality (2018-2023)

Higher values indicate stronger predictive power. Educational attainment shows strongest association, followed by median income. Model trained on 70% of data using 500 decision trees.

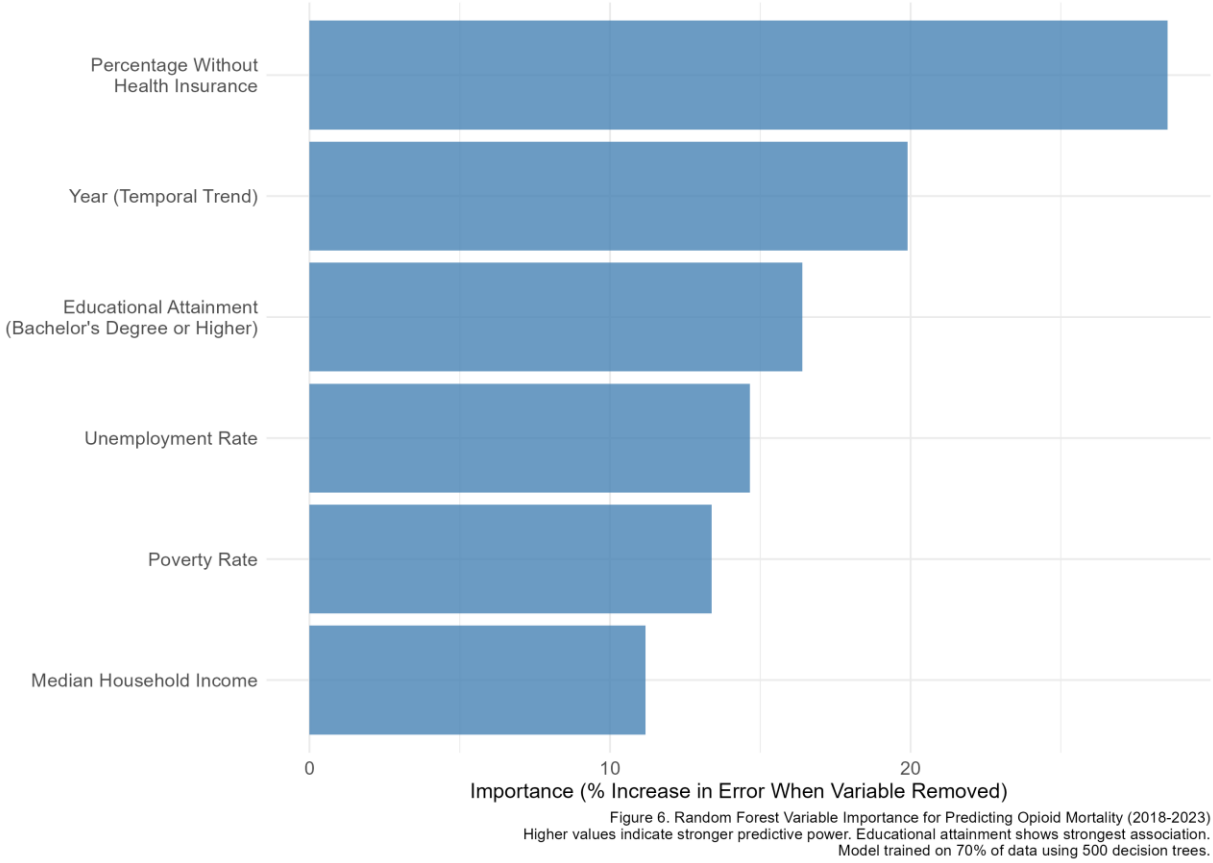

Figure 6. Random Forest Variable Importance for Predicting Opioid Mortality (2018-2023)